

Experiment 1: First Web API Using ASP.NET Core

Aim:

To create a simple ASP.NET Core Web API application and understand REST architecture, HTTP methods, HTTP status codes, and the structure of a Web API.

Objectives:

- Explain RESTful Web Service, Web API, and Microservice.
- Understand REST Architecture features.
- Learn HTTP Request and HTTP Response.
- Learn HTTP Action Verbs (GET, POST, PUT, DELETE).
- Learn HTTP Status Codes.
- Create a simple ASP.NET Core Web API.
- Understand the Web API project structure.
- Learn configuration files.

Software Requirements

- Visual Studio 2022 Community (Purple Icon)
- .NET 8 SDK
- Windows 10/11

Step 1: Install Visual Studio 2022

1. Download **Visual Studio 2022 Community**.
2. Run the installer.
3. Select the workload:

✓ **ASP.NET and web development**

4. Click Install.

Step 2: Open Visual Studio

Open **Visual Studio 2022**.

You'll see the start window.

Click

Create a new project

Step 3: Create a New Project

In the search box type

ASP.NET Core Web API

Select

ASP.NET Core Web API

Click

Next

Step 4: Configure the Project

Project Name

FirstWebAPI

Location

Choose any folder.

Solution Name

FirstWebAPI

Click

Next

Step 5: Additional Information

Select the following options carefully.

Framework

.NET 8.0 (Long Term Support)

Authentication Type

None

Check these options

✓ **Configure for HTTPS**

✓ **Use Controllers**

✓ **Enable OpenAPI Support**

Leave unchecked

Native AOT

Click

Create

Visual Studio will generate the project.

Step 6: Project Structure

The project will look like this.

FirstWebAPI

```
|
|  └─ Controllers
|      └─ WeatherForecastController.cs
|
|  └─ Properties
|      └─ launchSettings.json
|
|  └─ appsettings.json
|
|  └─ Program.cs
|
|  └─ WeatherForecast.cs
|
└─ FirstWebAPI.csproj
```

Understanding the Files

Controllers Folder

Contains all API Controllers.

Example

WeatherForecastController.cs

Program.cs

This is the entry point of the application.

It configures

- Services
 - Dependency Injection
 - Middleware
 - Routing
-

appsettings.json

Stores application settings.

Example

- Logging
 - Connection Strings
-

launchSettings.json

Stores

- Port Number
 - Launch URL
 - Environment
-

Step 7: Run the Project

Press

Ctrl + F5

If a dialog appears asking

Trust ASP.NET Core SSL Certificate?

Click

Yes

The browser opens

<https://localhost:7000/swagger>

You should see the Swagger page.

This means your project is working correctly.

Step 8: Delete Default Files

Delete

Controllers

WeatherForecastController.cs

Delete

WeatherForecast.cs

Step 9: Create a New Controller

Right-click

Controllers

Select

Add

↓

Class

Name

ValuesController.cs

Click

Add

Step 10: Write the Controller

Replace everything inside the file with the following code.

using Microsoft.AspNetCore.Mvc;

```
namespace FirstWebAPI.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class ValuesController : ControllerBase
    {
        [HttpGet]
        public IActionResult Get()
        {
            return Ok(new string[]
            {
                "Apple",
                "Banana",
                "Orange"
            });
        }

        [HttpPost]
        public IActionResult Post(string value)
        {
            return Ok("Received : " + value);
        }

        [HttpPut]
        public IActionResult Put(int id, string value)
        {
            return Ok($"Updated ID = {id}, Value = {value}");
        }
    }
}
```

```
        [HttpDelete]
        public IActionResult Delete(int id)
        {
            return Ok($"Deleted ID = {id}");
        }
    }
}
```

Save the file.

Step 11: Program.cs

Open

Program.cs

It should contain

```
var builder = WebApplication.CreateBuilder(args);
```

```
// Add services to the container.
```

```
builder.Services.AddControllers();
```

```
builder.Services.AddEndpointsApiExplorer();
```

```
builder.Services.AddSwaggerGen();
```

```
var app = builder.Build();
```

```
if (app.Environment.IsDevelopment())
```

```
{
```

```
    app.UseSwagger();
```

```
    app.UseSwaggerUI();
```

```
}
```

```
app.UseHttpsRedirection();
```

```
app.UseAuthorization();
```

```
app.MapControllers();
```

```
app.Run();
```

Step 12: Build the Project

Click

Build



Build Solution

Shortcut

Ctrl + Shift + B

If successful, you'll see

Build succeeded.

Step 13: Run the Project Again

Press

Ctrl + F5

Swagger opens.

Step 14: Test the GET Method

Click

GET

/api/Values

Click

Try it out

Click

Execute

Output

```
[  
  "Apple",  
  "Banana",  
  "Orange"  
]
```

Step 15: Test the POST Method

Click

POST

/api/Values

Enter

Laptop

Click

Execute

Output

Received : Laptop

Step 16: Test the PUT Method

Input

Id = 1

Value = Mobile

Output

Updated ID = 1, Value = Mobile

Step 17: Test the DELETE Method

Input

Id = 1

Output

Deleted ID = 1

REST Architecture

REST stands for **Representational State Transfer**.

Features:

- Client-Server Architecture
 - Stateless
 - Uniform Interface
 - Cacheable
 - Layered System
 - Uses HTTP Methods
 - Supports JSON and XML
-

What is a Web API?

A Web API is an interface that allows applications to communicate using HTTP requests and responses.

Example:

Client → HTTP Request → Web API → HTTP Response → Client

What is a Microservice?

A Microservice is a small independent service responsible for one business function.

Example:

- Student Service
 - Employee Service
 - Payment Service
 - Inventory Service
-

HTTP Request

A request sent from the client to the server.

Example

GET /api/values

Contains:

- URL
 - Method
 - Headers
 - Body (optional)
-

HTTP Response

A response sent from the server.

Example

```
[  
  "Apple",  
  "Banana",  
  "Orange"  
]
```

Contains:

- Status Code
- Headers

- Response Body

HTTP Action Verbs

Verb Purpose

GET Read data

POST Insert data

PUT Update data

DELETE Delete data

HTTP Status Codes

Code Meaning

200 OK

400 Bad Request

401 Unauthorized

500 Internal Server Error

Configuration Files

Program.cs

Configures services, middleware, routing, and dependency injection.

appsettings.json

Stores application configuration such as logging and connection strings.

launchSettings.json

Stores launch URLs, ports, and environment variables.

Startup.cs (.NET Core 3.1/.NET 5)

Used for application configuration. In .NET 6/7/8, its functionality is merged into Program.cs.

RouteConfig.cs (.NET Framework 4.5)

Defines URL routing.

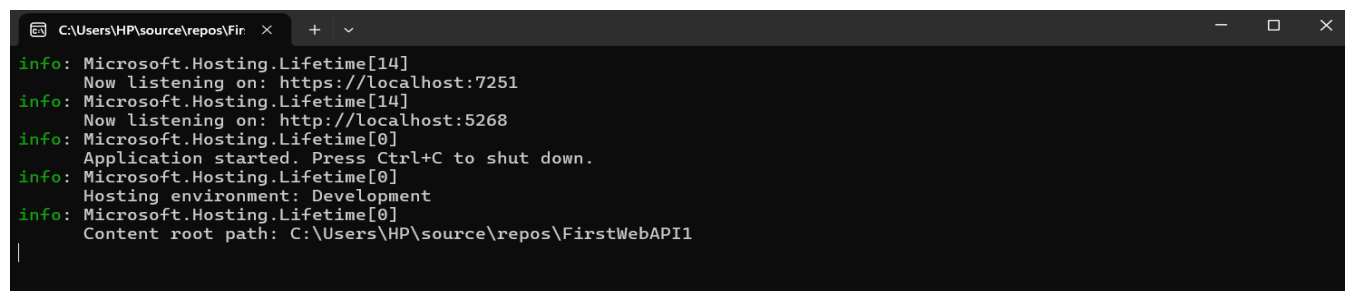
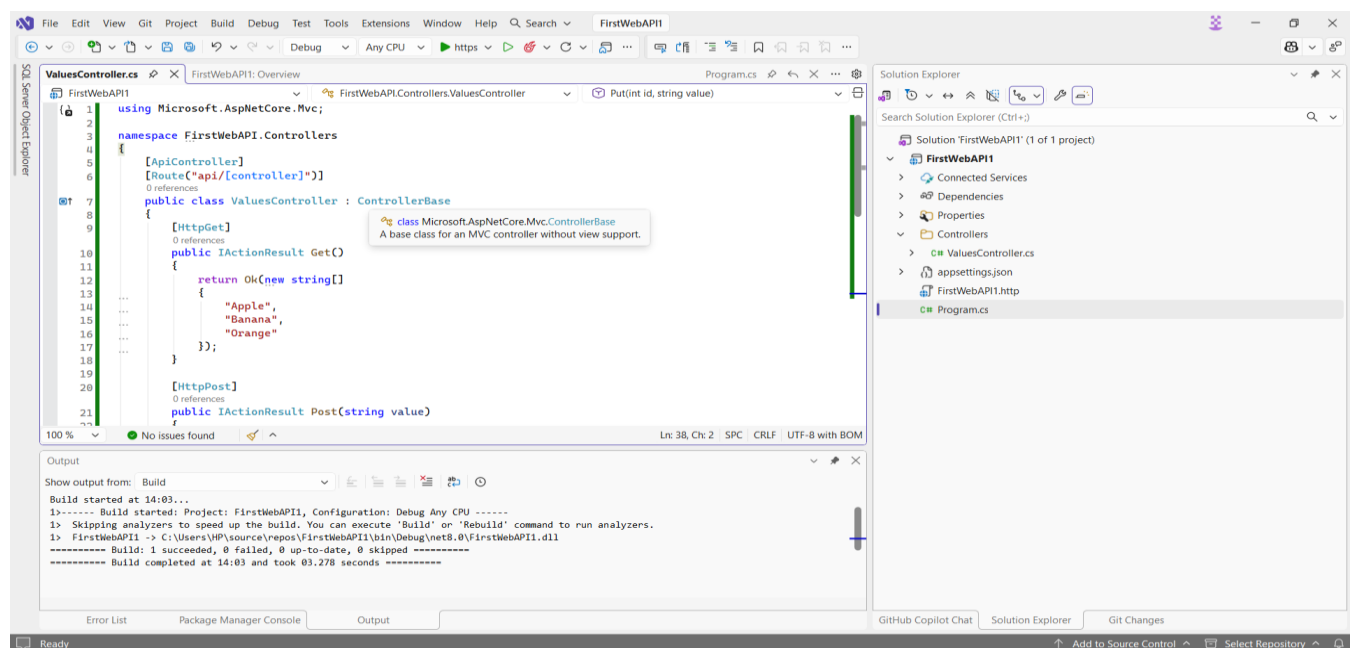
WebApiConfig.cs (.NET Framework 4.5)

Registers Web API routes.

Result

A simple ASP.NET Core Web API was created successfully. The API supports GET, POST, PUT, and DELETE operations and demonstrates REST architecture, HTTP requests and responses, action verbs, HTTP status codes, and the ASP.NET Core Web API project structure.

Outputs:



Microsoft Edge
It seems that your browser defaults had been changed unintentionally, and we have restored the default settings. [Learn more](#)

Swagger
Select a definition FirstWebAPI1 v1

FirstWebAPI1 ^{1.0} OAS 3.0

<https://localhost:7251/swagger/v1/swagger.json>

Values

- GET /api/Values
- POST /api/Values
- PUT /api/Values
- DELETE /api/Values

Microsoft Edge
It seems that your browser defaults had been changed unintentionally, and we have restored the default settings. [Learn more](#)

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7251/api/Values' \
  -H 'accept: */*'
```

Request URL

<https://localhost:7251/api/Values>

Server response

Code	Details
200	Response body <pre>{ "Apple", "Banana", "Orange" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Sat, 04 Jul 2026 10:26:16 GMT server: Kestrel</pre>

Responses

Code	Description	Links
200	OK	No links

Responses

Curl

```
curl -X 'DELETE' \
  https://localhost:7251/api/Values?id=1' \
  -H 'accept: */*'

```

Request URL

```
https://localhost:7251/api/Values?id=1

```

Server response

Code	Details
200	Response body Deleted 1 Response headers content-type: text/plain; charset=utf-8 date: Sat, 04 Jul 2026 18:28:14 GMT server: Kestrel

Responses

Code	Description	Links
200	OK	No links